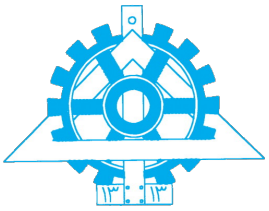


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پردیس دانشکده‌های فنی

دانشکده مهندسی برق و کامپیوتر



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Abstract

This assignment series explores fundamental concepts in probabilistic classification، parameter estimation، and statistical modeling. The exercises cover the derivation and comparison of Bayesian versus randomized decision rules، the computation of Bayes decision boundaries for Gaussian distributions، and Maximum Likelihood Estimation (MLE) for both discrete and continuous data. Additional tasks include modeling real-world count data with Poisson distributions، deriving the least squares loss from a probabilistic perspective، and studying the influence of class priors on decision boundaries. The latter sections apply Naive Bayes classifiers to synthetic and textual datasets، including spam detection، and develop interpretable feature-based classifiers for image data. Overall، the assignments aim to deepen understanding of probabilistic reasoning، decision theory، parameter estimation، and practical implementation of statistical models.

Question 1 – Bayes Rule vs. Random Decisions

We are tasked with classifying an input signal x into one of K classes $\omega \in \{1, 2, \ldots, K\}$، such that the action $\alpha(x) = i$ means classifying x into class i.

The Bayesian decision rule is to maximize the posterior probability:

$$\alpha_{\text{Bayes}}(x) = \omega^* = \arg\max_i p(\omega_i|x).$$

Now، suppose we replace it with a randomized decision rule، which classifies x into class i following the posterior probability $p(\omega_i|x)$، i.e.،

$$\alpha_{\text{rand}}(x) = \omega \sim p(\omega|x).$$

۱.۲ What is the overall risk R_{rand} for this decision rule؟

The overall risk for the randomized decision rule R_{rand} is the expected loss، where the loss is the zero-one loss function:

$$L(\alpha(x), \omega) = \begin{cases} 0 & \text{if} \alpha(x) = \omega \\ 1 & \text{if} \alpha(x) \neq \omega \end{cases}$$

Thus، the risk is:

$$R_{\text{rand}} = \mathbb{E}[L(\alpha_{\text{rand}}(x), \omega)] = \sum_{i=1}^{K} \int p(x) \cdot (1 - p(\omega_i|x))\, dx$$

۲.۲ Show that this risk R_{rand} is always no smaller than the Bayes risk R_{Bayes}.

The Bayes risk is the risk associated with the Bayesian decision rule، which always chooses the class with the highest posterior probability:

$$R_{\text{Bayes}} = \sum_{i=1}^{K} \int p(x) \cdot (1 - \delta_{i,\omega^*})\, dx$$

The Bayesian decision rule minimizes the risk by always choosing the class with the highest posterior probability. Since the randomized decision rule introduces uncertainty، its risk is always greater than or equal to the Bayes risk:

$$R_{\text{rand}} \geq R_{\text{Bayes}}$$

۳.۲ Under what conditions on the posterior are the two decision rules the same؟

The two decision rules will be the same when the posterior probabilities are deterministic، meaning for each x، one class has a posterior probability of ۱، and all other classes have posterior probability ۰:

$$p(\omega_i|x) = 1 \quad \text{class some for} \quad \omega_i \quad \text{and} \quad p(\omega_j|x) = 0 \quad \text{classes. other all for}$$

In this case، both the randomized and Bayesian decision rules will choose the same class deterministically.

Question 2 – Bayes Boundaries for Gaussian Classes

We now consider a two-class classification problem with salmon)ω_1(and sea bass)ω_2.(

۱.۳ One-dimensional Gaussian case

Assume the feature x is one-dimensional، and the class-conditional densities are Gaussian:

$$p(x|\omega_1) = \mathcal{N}(0, \sigma^2), \quad p(x|\omega_2) = \mathcal{N}(1, \sigma^2)$$

The Bayes decision rule chooses the class that minimizes the expected risk:

$$\text{Decide} \omega_1 \text{ if } \lambda_{21} P(\omega_2) p(x|\omega_2) \leq \lambda_{12} P(\omega_1) p(x|\omega_1)$$

and otherwise decide ω_2.

Using the Gaussian densities:

$$\frac{p(x|\omega_1)}{p(x|\omega_2)} = \exp\left(-\frac{(x-0)^2 - (x-1)^2}{2\sigma^2}\right) = \exp\left(\frac{2x-1}{2\sigma^2}\right)$$

Taking logarithms of both sides and solving for the threshold τ:

$$\frac{2\tau - 1}{2\sigma^2} = \ln \frac{\lambda_{12} P(\omega_2)}{\lambda_{21} P(\omega_1)}$$

$$\Rightarrow \tau = \frac{1}{2} - \sigma^2 \ln \frac{\lambda_{12} P(\omega_2)}{\lambda_{21} P(\omega_1)}$$

Thus، the threshold minimizing the average risk is:

$$\boxed{\tau = \frac{1}{2} - \sigma^2 \ln \frac{\lambda_{12} P(\omega_2)}{\lambda_{21} P(\omega_1)}}$$

When the loss values are symmetric)$\lambda_{12} = \lambda_{21} = 1$(and the priors are equal، the threshold simplifies to $\tau = 0.5$.

۲.۳ Two-dimensional Gaussian case

Now assume the feature vector is $x = [x_1, x_2]^T$، and that the two classes have the following parameters:

$$\mu_1 = \begin{bmatrix} 4 \\ 11 \end{bmatrix}, \quad \mu_2 = \begin{bmatrix} 10 \\ 3 \end{bmatrix}, \quad \Sigma = 3I, \quad P(\omega_1) = 0.6, \quad P(\omega_2) = 0.4$$

۱.۲.۳ (a) Discriminant functions

For Gaussian distributions with equal covariance matrices، the discriminant function is:

$$g_i(x) = -\frac{1}{2}(x - \mu_i)^T \Sigma^{-1} (x - \mu_i) + \ln P(\omega_i)$$

Therefore:

$$g_1(x) = -\frac{1}{2}(x - \mu_1)^T \Sigma^{-1} (x - \mu_1) + \ln 0.6$$

$$g_2(x) = -\frac{1}{2}(x - \mu_2)^T \Sigma^{-1} (x - \mu_2) + \ln 0.4$$

۲.۲.۳ (b) Decision boundary equation

The decision boundary is obtained by setting $g_1(x) = g_2(x)$:

$$(x - \mu_1)^T \Sigma^{-1} (x - \mu_1) - (x - \mu_2)^T \Sigma^{-1} (x - \mu_2) = 2 \ln \frac{P(\omega_1)}{P(\omega_2)}$$

Since $\Sigma = 3I \Rightarrow \Sigma^{-1} = \frac{1}{3} I$، this simplifies to:

$$(x - \mu_1)^T (x - \mu_1) - (x - \mu_2)^T (x - \mu_2) = 6 \ln \frac{0.6}{0.4}$$

which expands into a linear equation in x_1 and x_2.

۳.۲.۳ (c) Effect of priors and covariances

- Increasing $P(\omega_1)$ shifts the boundary toward ω_2، making classification more biased toward class ۱ (salmon).
- Decreasing $P(\omega_1)$ shifts the boundary toward ω_1، biasing toward class ۲ (sea bass).
- If the covariance matrices differ، the boundary becomes **quadratic** instead of linear، producing elliptical or hyperbolic separation curves.

۴.۲.۳ (d) Sampling and visualization

We can visualize the Bayes decision boundary by generating random samples from both classes and plotting the separating line using Python:

```
import numpy as np
import matplotlib.pyplot as plt


mu1 = np.array([4, 11])
mu2 = np.array([10, 3])
```

```
cov = np.array([[3, 0], [0, 3]])
inv_cov = np.linalg.inv(cov)
P1, P2 = 6.0, 4.0

# Generate random samples
class1_points = np.random.multivariate_normal(mu1, cov, 100)
class2_points = np.random.multivariate_normal(mu2, cov, 100)

# Decision boundary function
def decision_boundary(X, Y, mu1, mu2, inv_cov, P1, P2):
    xy = np.stack([X.ravel(), Y.ravel()])
    g1 = 5.0- * np.sum((xy.T - mu1) @ inv_cov * (xy.T - mu1), axis=1) +
    g2 = 5.0- * np.sum((xy.T - mu2) @ inv_cov * (xy.T - mu2), axis=1) +
    Z = g1 - g2
    return Z.reshape(X.shape)

# Grid and contour plot
x = np.linspace(0, 14, 300)
y = np.linspace(0, 14, 300)
X, Y = np.meshgrid(x, y)
Z = decision_boundary(X, Y, mu1, mu2, inv_cov, P1, P2)

plt.figure(figsize=(8, 6))
plt.scatter(class1_points[:,0], class1_points[:,1], color='blue', label
plt.scatter(class2_points[:,0], class2_points[:,1], color='red', label=
plt.contour(X, Y, Z, levels=[0], colors='green', linewidths=2)
plt.xlabel('Feature 1')
plt.ylabel('Feature 2')
plt.legend()
```

```
plt.title('Bayes Decision Boundary for Gaussian Classes')
plt.show()
```

This produces a two-dimensional plot showing samples from both Gaussian distributions and the green decision boundary $g_1(x) = g_2(x)$ that separates them.

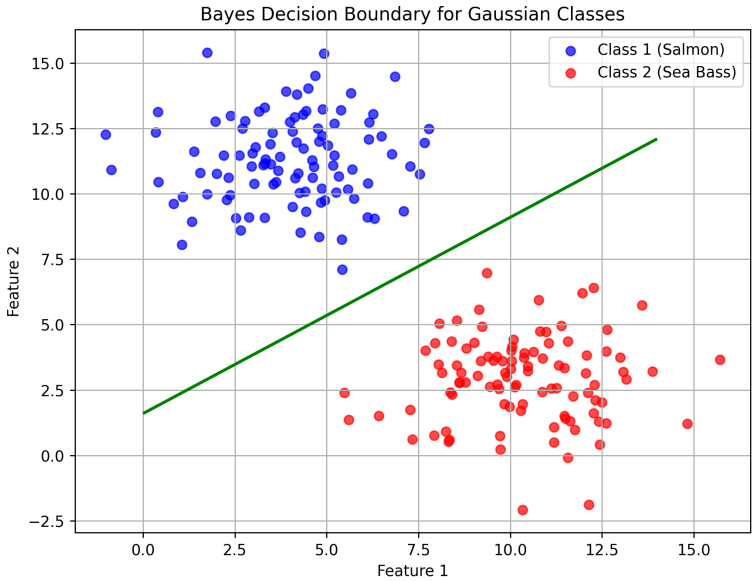


Figure : ١ Bayes Decision Boundary for Gaussian Classes

Question 3 – Naive Bayes Parameter Estimation (MLE)

The goal of this question is to find the Maximum Likelihood Estimates (MLEs) for the parameters $\phi = \{\phi_y, \phi_{j|y=0}, \phi_{j|y=1}\}$ by maximizing the joint log-likelihood $\ell(\phi)$. The model parameters are for the **Bernoulli distribution**:

- Prior probability: $\phi_y = P(y = 1)$
- Feature likelihoods: $\phi_{j|y=0} = P(x_j = 1|y = 0)$, $\phi_{j|y=1} = P(x_j = 1|y = 1)$

We will answer the following two parts:

.۱ Write the joint log-likelihood function $\ell(\phi)$.

.۲ Show that the MLEs correspond to the empirical frequencies.

۱.۴ Joint Log-Likelihood Function $\ell(\phi)$

Let N be the number of data points and d be the number of features. Denote the features for the i-th data point as $x_i = (x_{i1}, x_{i2}, \ldots, x_{id})$ and the class label as y_i. The likelihood of observing y_i and x_i is:

$$P(y_i, x_i) = P(y_i) \prod_{j=1}^{d} P(x_{ij}|y_i)$$

The log-likelihood function for N independent data points is the sum of the log of these likelihoods:

$$\ell(\phi) = \sum_{i=1}^{N} \log P(y_i, x_i)$$

Substituting the expressions for the class prior and feature likelihoods:

$$\ell(\phi) = \sum_{i=1}^{N} \log P(y_i) + \sum_{i=1}^{N} \sum_{j=1}^{d} \log P(x_{ij}|y_i)$$

The first term involves the class prior probabilities، and the second term involves the conditional feature probabilities. Expanding these terms:

$$\ell(\phi) = \sum_{i=1}^{N} \log \phi_{y_i} + \sum_{i=1}^{N} \sum_{j=1}^{d} \left[x_{ij} \log \phi_{j|y_i} + (1 - x_{ij}) \log(1 - \phi_{j|y_i}) \right]$$

Thus، the joint log-likelihood function is:

$$\boxed{\ell(\phi) = \sum_{i=1}^{N} \log \phi_{y_i} + \sum_{i=1}^{N} \sum_{j=1}^{d} \left[x_{ij} \log \phi_{j|y_i} + (1 - x_{ij}) \log(1 - \phi_{j|y_i}) \right]}$$

۲.۴ Maximum Likelihood Estimates (MLEs) Correspond to Empirical Frequencies

To find the MLEs، we need to differentiate the log-likelihood function $\ell(\phi)$ with respect to the parameters ϕ_y، $\phi_{j|y=0}$، and $\phi_{j|y=1}$، and then set the derivatives to zero.

۱.۲.۴ MLE for ϕ_y (Prior Probability)

The first part of the log-likelihood that depends on ϕ_y is:

$$\sum_{i=1}^{N} \log \phi_{y_i}$$

Differentiating with respect to ϕ_y، we get:

$$\frac{\partial \ell(\phi)}{\partial \phi_y} = \frac{N_y}{\phi_y}$$

where N_y is the number of instances where $y_i = y$، i.e.، $N_y = \sum_{i=1}^{N} \mathbb{I}(y_i = y)$.

Setting the derivative equal to zero and solving for ϕ_y، we obtain:

$$\hat{\phi}_y = \frac{N_y}{N}$$

Thus، the MLE for the class prior ϕ_y is simply the empirical frequency of class y in the data.

۲.۲.۴ MLE for $\phi_{j|y}$ (Feature Likelihoods)

Now، we differentiate the second part of the log-likelihood with respect to $\phi_{j|y}$. The relevant term is:

$$\sum_{i=1}^{N} \sum_{j=1}^{d} \left[x_{ij} \log \phi_{j|y_i} + (1 - x_{ij}) \log(1 - \phi_{j|y_i}) \right]$$

For each feature j and each class y، the derivative with respect to $\phi_{j|y}$ is:

$$\frac{\partial \ell(\phi)}{\partial \phi_{j|y}} = \sum_{i=1}^{N} \left[\frac{x_{ij}}{\phi_{j|y}} - \frac{1 - x_{ij}}{1 - \phi_{j|y}} \right]$$

Setting this equal to zero and solving for $\phi_{j|y}$، we get:

$$\hat{\phi}_{j|y} = \frac{\sum_{i=1}^{N} x_{ij} \mathbb{I}(y_i = y)}{\sum_{i=1}^{N} \mathbb{I}(y_i = y)}$$

Thus، the MLE for $\phi_{j|y}$ is the **empirical frequency** of feature j being ۱، given that the class label is y.

۳.۴ Conclusion

The MLEs for the Naive Bayes model correspond to the empirical frequencies of the class labels and feature values in the training data:

- The MLE for the class prior ϕ_y is the fraction of data points with class y.
- The MLE for the feature likelihood $\phi_{j|y}$ is the fraction of data points for which feature j is ۰،۱ given that the class label is y.

Thus، the MLEs provide a direct estimate of the parameters based on the observed frequencies in the data.

Question 4 – Modeling Goals with MLE

You are the Reign FC manager، and the team is five games into its ۲۰۲۱ season. The number of goals scored by the team in each game so far are given below:

$$x_1 = 2,\ x_2 = 4,\ x_3 = 6,\ x_4 = 0,\ x_5 = 1.$$

Let's call these scores $x_1, x_2, \ldots, x_5$. Based on your (assumed iid) data، you'd like to build a model to understand how many goals the Reign are likely to score in their next game. You decide to model the number of goals scored per game using a Poisson distribution. Recall that the Poisson distribution with parameter λ assigns every non-negative integer $x = 0, 1, 2, \ldots$ a probability given by

$$\text{Poi}(x|\lambda) = \frac{e^{-\lambda}\lambda^x}{x!}.$$

۱.۵ Maximum-Likelihood Estimate for λ

The likelihood function for a Poisson distribution، given the goal counts $x_1, x_2, \ldots, x_n$ from n games، is the product of the individual probabilities for each game:

$$L(\lambda) = \prod_{i=1}^{n} \frac{e^{-\lambda}\lambda^{x_i}}{x_i!}.$$

Taking the natural logarithm of the likelihood function to obtain the log-likelihood function:

$$\ell(\lambda) = \sum_{i=1}^{n} \log\left(\frac{e^{-\lambda}\lambda^{x_i}}{x_i!}\right) = \sum_{i=1}^{n} \left(-\lambda + x_i \log(\lambda) - \log(x_i!)\right).$$

Simplifying the expression:

$$\ell(\lambda) = -n\lambda + \log(\lambda) \sum_{i=1}^{n} x_i - \sum_{i=1}^{n} \log(x_i!).$$

Note that the last sum does not depend on λ، so we can ignore it when maximizing the log-likelihood with respect to λ. Thus، the simplified log-likelihood function is:

$$\ell(\lambda) = -n\lambda + \log(\lambda) \sum_{i=1}^{n} x_i.$$

To find the maximum likelihood estimate، we take the derivative of $\ell(\lambda)$ with respect to λ and set it to zero:

$$\frac{d\ell(\lambda)}{d\lambda} = -n + \frac{\sum_{i=1}^{n} x_i}{\lambda}.$$

Setting this equal to zero:

$$-n + \frac{\sum_{i=1}^{n} x_i}{\lambda} = 0,$$

$$\hat{\lambda} = \frac{\sum_{i=1}^{n} x_i}{n}.$$

Thus، the maximum likelihood estimate of λ is the sample mean of the goal counts $x_1, x_2, \ldots, x_n$.

۲.۵ Numerical Estimate of λ After the First Five Games

Given the goal counts for the first five games: $x_1 = 2, x_2 = 4, x_3 = 6, x_4 = 0, x_5 = 1$، the sum of the goal counts is:

$$\sum_{i=1}^{5} x_i = 2 + 4 + 6 + 0 + 1 = 13.$$

The number of games is $n = 5$، so the MLE for λ is:

$$\hat{\lambda} = \frac{13}{5} = 2.6.$$

Thus، the estimated value of λ after the first five games is $\hat{\lambda} = 2.6$.

To compute the probability that the Reign score ۶ goals in their next game، we use the Poisson probability mass function:

$$P(x = 6|\lambda) = \frac{e^{-\lambda}\lambda^x}{x!}.$$

Substituting $\lambda = 2.6$ and $x = 6$:

$$P(x = 6|2.6) = \frac{e^{-2.6}2.6^6}{6!}.$$

First، compute $6! = 720$، and then:

$$P(x = 6|2.6) = \frac{e^{-2.6}2.6^6}{720}.$$

Using a calculator:

$$P(x = 6|2.6) \approx \frac{0.074273 \times 107.3741824}{720} \approx \frac{7.976}{720} \approx 0.01108.$$

Thus، the probability that the Reign score ۶ goals in their next game is approximately 0.01108 or ۱.۱۱%.

۳.۵ Updated Estimate of λ After the Sixth Game

Suppose the Reign score ۸ goals in their ۶th game. The updated data set is now:

$$x_1 = 2, x_2 = 4, x_3 = 6, x_4 = 0, x_5 = 1, x_6 = 8.$$

The sum of the goal counts is now:

$$\sum_{i=1}^{6} x_i = 2 + 4 + 6 + 0 + 1 + 8 = 21.$$

The number of games is now $n = 6$، so the updated MLE for λ is:

$$\hat{\lambda} = \frac{21}{6} = 3.5.$$

Thus، the updated value of λ after the ۶th game is $\hat{\lambda} = 3.5$.

To compute the probability that the Reign score ۶ goals in their ۷th game، we again use the Poisson probability mass function:

$$P(x = 6|\lambda = 3.5) = \frac{e^{-3.5}3.5^6}{6!}.$$

Substituting $\lambda = 3.5$ and $x = 6$:

$$P(x = 6|3.5) = \frac{e^{-3.5}3.5^6}{720}.$$

Using a calculator:

$$P(x = 6|3.5) \approx \frac{0.030197 \times 1838.265625}{720} \approx \frac{55.485}{720} \approx 0.0770.$$

Thus، the probability that the Reign score ۶ goals in their ۷th game is approximately 0.0770 or ۷٫۷۰%.

Question 5 – MLE Derivation of the Least Squares Loss

In Linear Regression، we assume the relationship between input $x^{(i)}$ and output $y^{(i)}$ is

$$y^{(i)} = \theta^T x^{(i)} + \epsilon^{(i)},$$

where the error $\epsilon^{(i)}$ is independent and follows a Gaussian (Normal) distribution with mean zero and variance σ^2، i.e.،

$$\epsilon^{(i)} \sim \mathcal{N}(0, \sigma^2).$$

۱.۶ Likelihood Function $p(y^{(i)}|x^{(i)}; \theta, \sigma^2)$

Given the model $y^{(i)} = \theta^T x^{(i)} + \epsilon^{(i)}$، the error term $\epsilon^{(i)}$ follows a Gaussian distribution، i.e.، $\epsilon^{(i)} \sim \mathcal{N}(0, \sigma^2)$. Therefore، the conditional distribution of $y^{(i)}$ given $x^{(i)}$، θ، and σ^2 is also Gaussian:

$$p(y^{(i)}|x^{(i)}; \theta, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y^{(i)} - \theta^T x^{(i)})^2}{2\sigma^2}\right).$$

Thus، the likelihood for a single data point $(x^{(i)}, y^{(i)})$ is given by the above expression. For a dataset of n independent observations $(x^{(i)}, y^{(i)})$، the joint likelihood is the product of the individual likelihoods:

$$L(\theta, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y^{(i)} - \theta^T x^{(i)})^2}{2\sigma^2}\right).$$

۲.۶ Log-Likelihood and Maximization with Respect to θ

To simplify the maximization process، we take the natural logarithm of the likelihood function to obtain the log-likelihood:

$$\ell(\theta, \sigma^2) = \log L(\theta, \sigma^2) = \sum_{i=1}^{n} \left[-\frac{1}{2} \log(2\pi\sigma^2) - \frac{(y^{(i)} - \theta^T x^{(i)})^2}{2\sigma^2} \right].$$

Since the term $-\frac{1}{2}\log(2\pi\sigma^2)$ does not depend on θ، we can ignore it when maximizing the log-likelihood with respect to θ. Therefore، the log-likelihood simplifies to:

$$\ell(\theta) = -\frac{1}{2\sigma^2} \sum_{i=1}^{n} \left(y^{(i)} - \theta^T x^{(i)} \right)^2.$$

To maximize this log-likelihood with respect to θ، we take the derivative of $\ell(\theta)$ with respect to θ and set it to zero:

$$\frac{\partial \ell(\theta)}{\partial \theta} = \frac{1}{\sigma^2} \sum_{i=1}^{n} \left(y^{(i)} - \theta^T x^{(i)} \right) x^{(i)} = 0.$$

This leads to the normal equation:

$$\sum_{i=1}^{n} \left(y^{(i)} - \theta^T x^{(i)} \right) x^{(i)} = 0.$$

Expanding the equation and solving for θ، we obtain:

$$\hat{\theta} = \left(\sum_{i=1}^{n} x^{(i)} (x^{(i)})^T \right)^{-1} \sum_{i=1}^{n} x^{(i)} y^{(i)}.$$

Thus، the maximum likelihood estimate for θ corresponds to solving this normal equation، which is equivalent to minimizing the sum of squared errors.

۳.۶ Connection to the Mean Squared Error (MSE)

The log-likelihood maximization problem we have just solved is equivalent to minimizing the following least squares cost function (mean squared error):

$$\text{MSE}(\theta) = \frac{1}{n} \sum_{i=1}^{n} \left(y^{(i)} - \theta^T x^{(i)} \right)^2 .$$

Multiplying both sides of the MSE by n، we get:

$$\text{MSE}(\theta) = \frac{1}{n} \sum_{i=1}^{n} \left(y^{(i)} - \theta^T x^{(i)} \right)^2 .$$

Thus، the log-likelihood maximization process results in the same solution as minimizing the squared error cost function. Therefore، maximizing the log-likelihood is equivalent to minimizing the Mean Squared Error (MSE)، which is a standard loss function used in linear regression.

Question 6 – Effect of Priors on the Decision Boundary

Consider a two-class، one-dimensional classification problem with the classes ω_1 and ω_2، where the likelihoods are standard Gaussians with equal variance σ^2 but different means:

$$p(x|\omega_1) \sim \mathcal{N}(\mu_1, \sigma^2) \quad \text{and} \quad p(x|\omega_2) \sim \mathcal{N}(\mu_2, \sigma^2).$$

We are tasked with deriving the decision boundary condition x^* when the priors $P(\omega_1)$ and $P(\omega_2)$ are unequal. Then، we will discuss how the decision boundary shifts as the prior probability of ω_1، $P(\omega_1)$، increases relative to $P(\omega_2)$.

١.٧ Derivation of the Decision Boundary Condition

The decision boundary occurs when the posterior probabilities for both classes are equal. Using Bayes' theorem، we can write the posterior probability for each class as:

$$P(\omega_1|x) = \frac{P(x|\omega_1)P(\omega_1)}{P(x)}, \quad P(\omega_2|x) = \frac{P(x|\omega_2)P(\omega_2)}{P(x)}.$$

The decision rule is to classify x as ω_1 if $P(\omega_1|x) > P(\omega_2|x)$، and classify x as ω_2 if $P(\omega_2|x) > P(\omega_1|x)$.

Since the denominator $P(x)$ is the same for both posteriors، we can ignore it and instead compare the numerators:

$$P(x|\omega_1)P(\omega_1) \overset{?}{>} P(x|\omega_2)P(\omega_2).$$

Taking the natural logarithm of both sides to simplify، we get:

$$\log P(x|\omega_1) + \log P(\omega_1) \overset{?}{>} \log P(x|\omega_2) + \log P(\omega_2).$$

Now، since the likelihoods $P(x|\omega_1)$ and $P(x|\omega_2)$ are Gaussian، we substitute the Gaussian probability density functions:

$$P(x|\omega_1) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu_1)^2}{2\sigma^2}\right), \quad P(x|\omega_2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu_2)^2}{2\sigma^2}\right).$$

Substituting these into the inequality:

$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu_1)^2}{2\sigma^2}\right) P(\omega_1) \overset{?}{>} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu_2)^2}{2\sigma^2}\right) P(\omega_2).$$

The terms $\frac{1}{\sqrt{2\pi\sigma^2}}$ cancel out، and we are left with:

$$\exp\left(-\frac{(x-\mu_1)^2}{2\sigma^2}\right) P(\omega_1) \overset{?}{>} \exp\left(-\frac{(x-\mu_2)^2}{2\sigma^2}\right) P(\omega_2).$$

Taking the natural logarithm of both sides again:

$$-\frac{(x-\mu_1)^2}{2\sigma^2} + \log P(\omega_1) \overset{?}{>} -\frac{(x-\mu_2)^2}{2\sigma^2} + \log P(\omega_2).$$

Simplifying:

$$\frac{(x-\mu_2)^2 - (x-\mu_1)^2}{2\sigma^2} \overset{?}{<} \log \frac{P(\omega_1)}{P(\omega_2)}.$$

Expanding the terms in the numerator:

$$\frac{(x^2 - 2x\mu_2 + \mu_2^2) - (x^2 - 2x\mu_1 + \mu_1^2)}{2\sigma^2} = \frac{-2x(\mu_2 - \mu_1) + (\mu_2^2 - \mu_1^2)}{2\sigma^2}.$$

This simplifies to the decision boundary condition:

$$x^* = \frac{\mu_1 + \mu_2}{2} - \frac{\sigma^2}{\mu_2 - \mu_1} \log \frac{P(\omega_1)}{P(\omega_2)}.$$

۲.۷ Effect of Priors on the Decision Boundary

From the derived decision boundary equation، we see that the position of the decision boundary depends on the prior probabilities $P(\omega_1)$ and $P(\omega_2)$. Specifically، as the ratio $\frac{P(\omega_1)}{P(\omega_2)}$ increases (i.e.، as the prior probability of ω_1 increases relative to ω_2،(the decision boundary x^* shifts towards μ_1، the mean of class ω_1.

Thus، increasing the prior probability of ω_1 makes it more likely that we will classify an observation as belonging to ω_1، even if x is closer to μ_2، and this shifts the boundary away from μ_2 towards μ_1. Similarly، increasing the prior probability of ω_2 will shift the boundary towards μ_2.

This demonstrates how the decision boundary is influenced by the relative priors of the classes. Larger prior probabilities for one class lead to a decision boundary that favors that class.

Question 7 — Naive Bayes Warm-Up

This notebook demonstrates two main parts:

١. Using Bayes' theorem to compute a posterior probability in a simple drug test example.

٢. Implementing a Naive Bayes classifier for a small binary dataset، showing both the unsmoothed (MLE) and Laplace-smoothed versions.

١.٨ Part ١ — Drug Test Probability Example

We are given the following probabilities:

$$P(T = 1 \mid D = 1) = 0.99, \quad P(T = 0 \mid D = 0) = 0.99, \quad P(D = 1) = 0.001.$$

From this، we have:

$$P(T = 1 \mid D = 0) = 0.01, \quad P(D = 0) = 1 - P(D = 1) = 0.999.$$

We want to compute:

$$P(D = 1 \mid T = 1) = \frac{P(T = 1 \mid D = 1)P(D = 1)}{P(T = 1)},$$

where

$$P(T = 1) = P(T = 1 \mid D = 1)P(D = 1) + P(T = 1 \mid D = 0)P(D = 0).$$

Substituting the given values:

$$P(T=1) = (0.99)(0.001) + (0.01)(0.999) = 0.01098,$$
$$P(D=1 \mid T=1) = \frac{0.99 \times 0.001}{0.01098} = 0.09016.$$

Thus، even though the test is highly accurate، the posterior probability that a person actually has the disease given a positive test result is only about 9.0%. This result highlights the importance of base rates in probabilistic reasoning.

۲.۸ Part ۲ — Naive Bayes Classifier

We are given ۱۰ examples with two binary features:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 0 & 0 \\ 1 & 1 \\ 1 & 1 \\ 0 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}, \quad y = \begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}^T.$$

We want to classify the test example:

$$x = [1, 1].$$

Step ۱ — MLE (Unsmoothed) Naive Bayes

The Maximum Likelihood Estimators (MLE) for Bernoulli Naive Bayes are:

$$\hat{P}(y=c) = \frac{N_c}{N}, \quad \hat{P}(x_j = 1 \mid y = c) = \frac{N_{x_j=1, y=c}}{N_{y=c}},$$

where N_c is the number of examples with label c.

However، if a feature never appears with a given class، the denominator causes zero probabilities—leading to numerical issues.

From the data:

$$\begin{aligned} P(y=1) = 0.5, \quad P(y=0) = 0.5, \\ P(x_1 = 1 \mid y = 1) = 0.8, \quad P(x_2 = 1 \mid y = 1) = 1.0, \\ P(x_1 = 1 \mid y = 0) = 0.6, \quad P(x_2 = 1 \mid y = 0) = 0.0. \end{aligned}$$

The posterior probabilities for $x = [1, 1]$ are:

$$\begin{aligned} P(y=1 \mid x) \propto P(y=1)\, P(x_1 = 1 \mid y = 1)\, P(x_2 = 1 \mid y = 1), \\ P(y=0 \mid x) \propto P(y=0)\, P(x_1 = 1 \mid y = 0)\, P(x_2 = 1 \mid y = 0). \end{aligned}$$

Because $P(x_2 = 1 \mid y = 0) = 0$، we get:

$$P(y=1 \mid x) = 1.0000, \quad P(y=0 \mid x) = 0.0000,$$

and therefore the predicted class is $y = 1$.

Step ۲ — Laplace-Smoothing (Add-۱)

To avoid zero probabilities، we apply Laplace (add-one) smoothing:

$$\tilde{P}(x_j = 1 \mid y = c) = \frac{N_{x_j=1, y=c} + 1}{N_{y=c} + 2}.$$

This ensures that no feature-class probability is ever exactly zero.

The smoothed conditional probabilities become:

$$P(x_1 = 1 \mid y = 1) = 0.714, \quad P(x_2 = 1 \mid y = 1) = 0.857,$$
$$P(x_1 = 1 \mid y = 0) = 0.571, \quad P(x_2 = 1 \mid y = 0) = 0.143.$$

The smoothed posterior probabilities for $x = [1, 1]$ are:

$$P(y = 1 \mid x) = 0.8824, \quad P(y = 0 \mid x) = 0.1176,$$

thus again predicting $y = 1$، but with more realistic probabilities.

۳.۸ Summary and Comparison

Version	$P(x_2 = 1 \mid y = 0)$	$P(y = 1 \mid x)$	**Predicted Class**
MLE (Unsmoothed)	۰.۰۰۰	۱.۰۰۰۰	۱
Laplace-Smoothed	۰.۱۴۳	۰.۸۸۲۴	۱

۴.۸ Conclusions

Without smoothing، unseen feature-class combinations cause zero probabilities، which make the model overconfident. Laplace smoothing ensures stable، nonzero estimates and more calibrated posterior probabilities. This technique is the foundation of the Naive Bayes classifier used in text classification and many domains with sparse binary features.

Question 8 — Classifying Spam with Naive Bayes

۱.۹ Overview

In this question، we implement a Naive Bayes classifier to distinguish between *spam* and *ham* emails. Each word is treated as a binary or count-based feature، assuming conditional independence given the class label.

The main steps are as follows:

۱. Load and preprocess the training/test data.

۲. Compute prior probabilities.

۳. Estimate conditional probabilities using the m-estimate method.

۴. Classify test emails and compute accuracy.

۵. Study the effect of different m values.

Each line in the dataset is parsed into a dictionary containing the email ID، label، and a mapping of word counts.

۲.۹ Load Dataset

We read and parse the emails from text files containing word counts per message. A total of 9000 training and 1000 test emails were loaded.

۳.۹ Compute Prior Probabilities

We estimate the prior probabilities using Maximum Likelihood Estimation (MLE):

$$P(\text{spam}) = \frac{N_{\text{spam}}}{N_{\text{total}}}, \quad P(\text{ham}) = \frac{N_{\text{ham}}}{N_{\text{total}}}.$$

From the dataset:

$$P(\text{spam}) = 0.5737, \quad P(\text{ham}) = 0.4263.$$

۴.۹ Vocabulary and Word Counts

We collect all unique words from the training set to form the vocabulary V. Then، we count how many times each word appears in spam and ham messages.

$$|V| = 1330.$$

Let:

$$\text{spam_counts}(w) = \text{ word times of number} w \text{spam in appears}$$

$$\text{ham_counts}(w) = \text{ word times of number} w \text{ham in appears .}$$

۵.۹ Conditional Probabilities with m-Estimate

We estimate the conditional probabilities using the m-estimate formula:

$$P(w_i \mid c) = \frac{n_i + mp}{N + m},$$

where:

- n_i = count of word w_i in class c،
- N = total word count in class c،
- $p = \frac{1}{|V|}$ = prior probability estimate،
- m = strength of prior (smoothing parameter).

Using $m = |V|$، the top ۵ words for each class are:

words: spam Top{... “smtpsv”، “smtprs”، ،”۲۳۷“ “corp”،}

words: ham Top{... ،”۱“ ،”۳“ ،”۲“ ،”۴“}

۶.۹ Classifying Test Emails

We use log-probabilities to prevent numerical underflow. An email is classified as spam if:

$$\log P(\text{spam} \mid x) > \log P(\text{ham} \mid x),$$

where:

$$\log P(c \mid x) \propto \log P(c) + \sum_i x_i \log P(w_i \mid c).$$

The test accuracy obtained for $m = |V|$ is:

$$\text{Accuracy} = 0.9290.$$

۷.۹ Varying m and Observing the Effect

We vary the smoothing parameter m to examine its effect on accuracy:

$$m = |V| \times [1, 10, 100, 1000, 10000].$$

The results are summarized below:

m	**Accuracy**
1330	0.9290
13300	0.9240
133000	0.9090
1330000	0.8500
13300000	0.8370

Table :۱ Effect of different m values on test accuracy.

٨.٩ Accuracy vs. Smoothing Parameter m

The figure below shows how the test accuracy changes as we vary the smoothing parameter m:

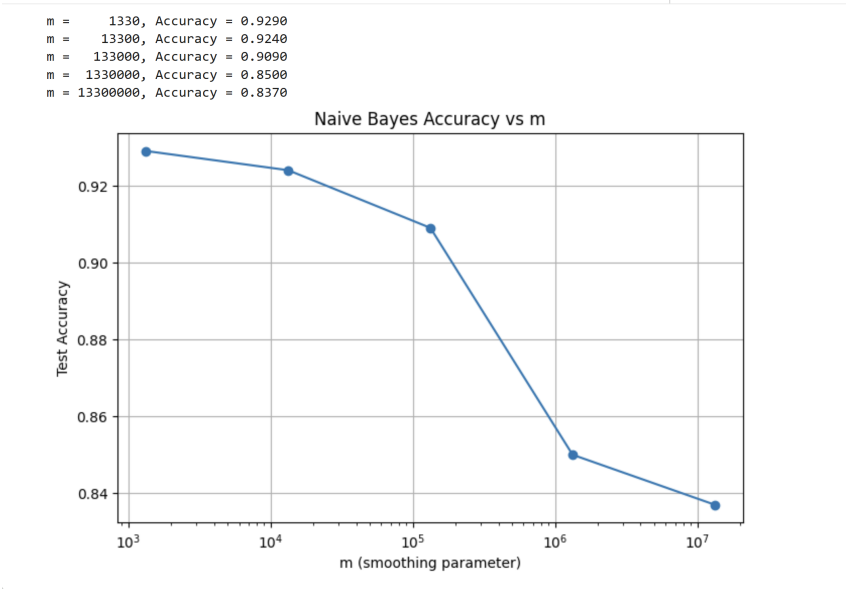


Figure : ٢ Effect of different m values on test accuracy.

As m increases، the model becomes less data-driven and more influenced by the uniform prior. Accuracy decreases gradually as m becomes excessively large.

٩.٩ Conclusion

- When m is small، the model relies heavily on the observed data — leading to possible overfitting.
- When m is large، the model's conditional probabilities approach the uniform prior — reducing overfitting but possibly causing underfitting.
- The best accuracy is generally achieved with moderate m values.

This exercise demonstrates the practical use of the m-estimate method

to smooth Naive Bayes probabilities and balance the bias–variance tradeoff in text classification.

Question 9 – Cloud vs. Clear Sky Classification

In this problem، we aim to design an classifier to distinguish between cloudy and clear-sky images. The dataset contains ۴۰ cloudy $(c_1, c_2, \ldots, c_{40})$ and ۴۰ clear-sky $(s_1, s_2, \ldots, s_{40})$ images. No pre-trained or well-known machine learning models are used؛ instead، we construct simple، explainable criteria derived from observable image features.

۱.۱۰ Dataset Partitioning

Following standard practice، the dataset was divided as:

$$\text{Training Set: } 60\% \quad \text{and} \quad \text{Test Set: } 40\%.$$

Thus، ۴۸ images (۲۴ cloudy + ۲۴ clear) were used for training، and ۳۲ images (۱۶ cloudy + ۱۶ clear) for testing.

۲.۱۰ Feature Definition

We define four interpretable quantitative features derived from each image's RGB pixel values and intensity structure.

(a) Blueness Ratio

Clear skies tend to have a higher proportion of blue pixels compared to cloudy skies. For each pixel، we define:

$$B_{\text{ratio}} = \frac{N(B > \frac{R+G}{2})}{N_{\text{total}}},$$

where R, G, B denote the normalized red، green، and blue intensities، and N_{total} is the total number of pixels.

(b) Brightness

Cloudy skies generally exhibit lower global brightness due to light scattering. The average normalized brightness of the image is defined as:

$$\text{Brightness} = \frac{1}{N_{\text{total}}} \sum_{i=1}^{N_{\text{total}}} \frac{R_i + G_i + B_i}{3}.$$

(c) Edge Density

Cloud edges and sunlight reflections both introduce texture، but cloudy skies may show less sharp structural contrast. We quantify structural complexity using the proportion of detected edges:

$$E_{\text{density}} = \frac{N(\text{Canny by detected edges})}{N_{\text{total}}}.$$

(d) Sun Tint (Orangish Component)

In clear or sunny conditions، the presence of sunlight often introduces orange or yellow tints in parts of the image. We measure the proportion of such pixels as:

$$S_{\text{tint}} = \frac{N(R > 1.2G \ \wedge \ R > 1.2B)}{N_{\text{total}}}.$$

۳.۱۰ Threshold Selection via Histogram Analysis

For each feature، histograms of cloudy and clear training images were plotted to observe separability. The threshold for each feature was set at the midpoint between class means:

$$T_f = \frac{\mu_{\text{cloudy},f} + \mu_{\text{clear},f}}{2}, \quad f \in \{\text{blueness}, \text{brightness}, \text{density edge}, \text{tint sun}\}.$$

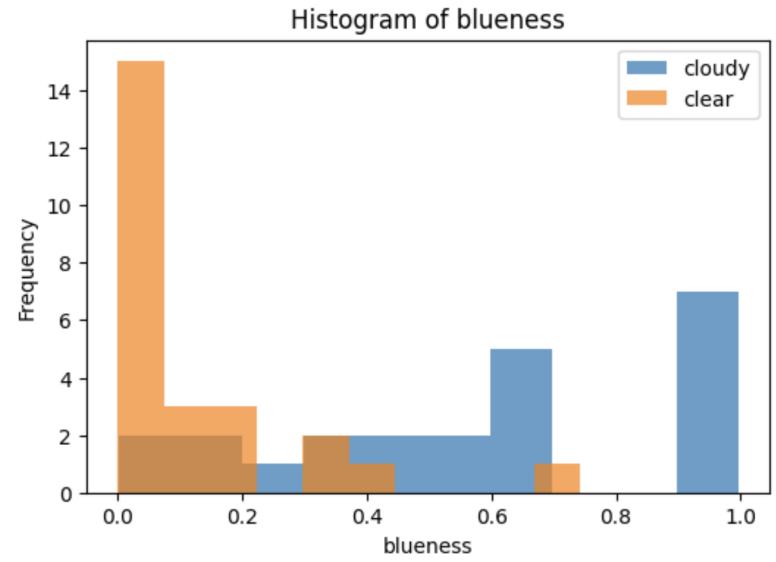
These thresholds allow each feature to independently classify an image by a simple rule.

۴.۱۰ Classification Rules

Each feature provides an interpretable binary decision:

$$\begin{aligned}
&\text{if Cloudy} \quad B_{\text{ratio}} > T_{\text{blueness}}, \\
&\text{if Cloudy} \quad \text{Brightness} < T_{\text{brightness}}, \\
&\text{if Cloudy} \quad E_{\text{density}} > T_{\text{edge}}, \\
&\text{if Cloudy} \quad S_{\text{tint}} < T_{\text{tint}}.
\end{aligned}$$

۵.۱۰ Feature Histograms and Individual Classifier Performance




```
blueness threshold set to 0.346
```

Figure :۳ Histogram of Blueness Ratio for cloudy and clear images

$$\text{Confusion Matrix} = \begin{bmatrix} 11 & 6 \\ 4 & 11 \end{bmatrix}, \quad \text{Precision} = 0.733, \quad \text{Recall} = 0.647$$

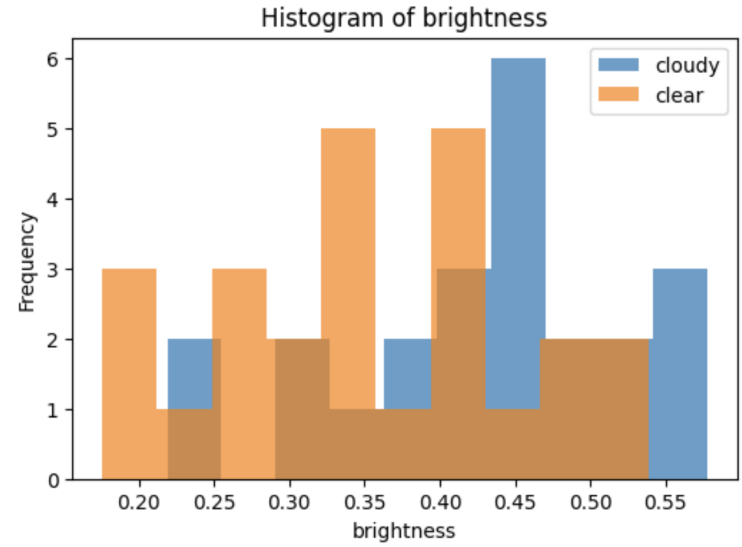


Figure :۴ Histogram of Brightness for cloudy and clear images

$$\text{Confusion Matrix} = \begin{bmatrix} 8 & 9 \\ 8 & 7 \end{bmatrix}, \quad \text{Precision} = 0.500, \quad \text{Recall} = 0.471$$

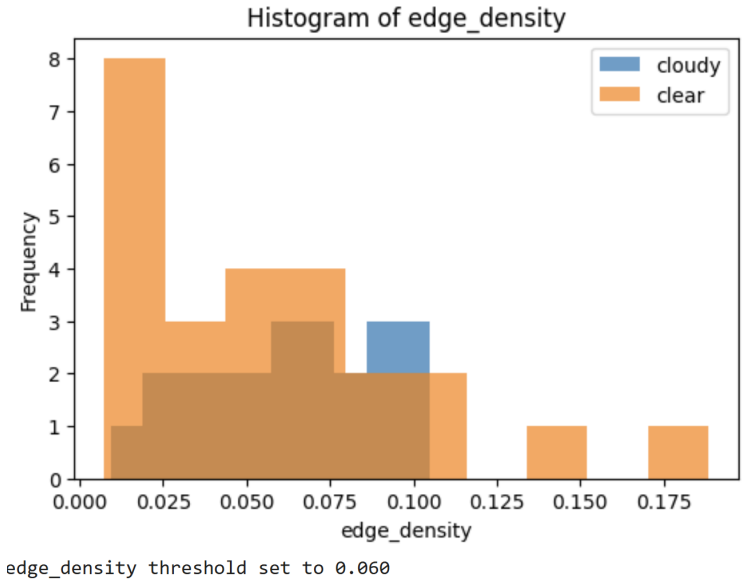


Figure :۵ Histogram of Edge Density for cloudy and clear images

$$\text{Confusion Matrix} = \begin{bmatrix} 5 & 12 \\ 7 & 8 \end{bmatrix}, \quad \text{Precision} = 0.417, \quad \text{Recall} = 0.294$$

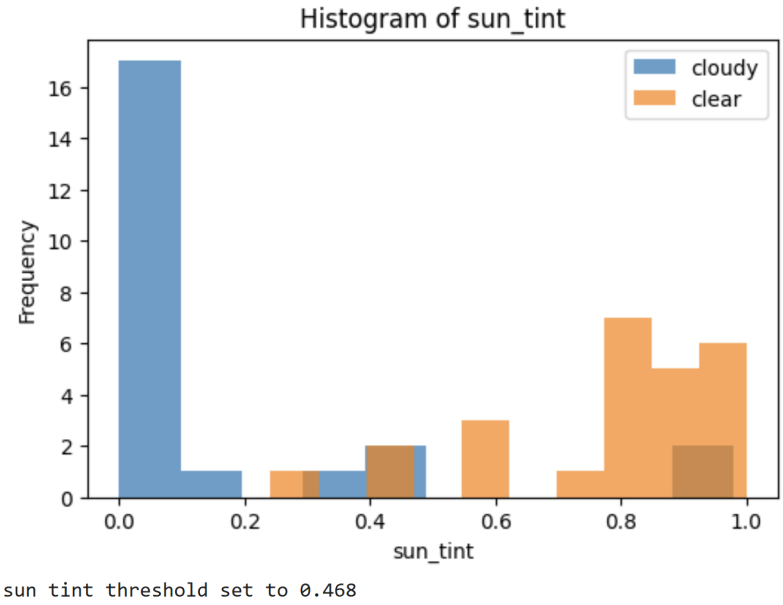


Figure :۶ Histogram of Sun Tint for cloudy and clear images

$$\text{Confusion Matrix} = \begin{bmatrix} 13 & 4 \\ 3 & 12 \end{bmatrix}, \quad \text{Precision} = 0.812, \quad \text{Recall} = 0.765$$

۶.۱۰ Combined Criterion (Majority Voting)

To improve robustness، we combine the four binary rules into a single classifier using majority voting:

$$\text{Label(image)} = \begin{cases} \text{Cloudy}, & \text{true are above conditions ۴ of ۲ least at if,} \\ \text{Clear}, & \text{otherwise.} \end{cases}$$

Combined classifier results:

$$\text{Matrix Confusion} = \begin{bmatrix} 12 & 5 \\ 6 & 9 \end{bmatrix}, \quad \text{Precision} = 0.667, \quad \text{Recall} = 0.706$$

۷.۱۰ Discussion of Misclassifications

Misclassified images were visually inspected. Common causes include:

- Bright cloudy images with sunlight were misclassified as clear due to high brightness or sun-tint.
- Very blue-tinted cloudy scenes (e.g.، thin clouds or post-rain conditions) were wrongly predicted as clear due to elevated blueness ratio.
- Some clear images with tree or building edges were misclassified as cloudy because of high edge density.

۸.۱۰ Conclusion

Among the individual criteria، brightness and sun-tint provided more stable separation between classes. Edge detection performed worst because both cloudy and sunny scenes can produce strong edge responses—cloud contours and sunlight reflections both trigger edges، leading to confusion. Similarly، color-based features may fail when clouds are illuminated by sunlight or when atmospheric scattering causes blue light to dominate even in cloudy scenes.

Overall، combining all four interpretable criteria through majority voting yields a balanced and explainable decision rule that can separate cloudy and clear skies without requiring complex models.